Strategic Data Source Discovery for the "Fessi" Knowledge Graph: Batteries & Accumulators

Executive Summary

The effective management of waste streams within a university setting requires a sophisticated data architecture capable of navigating complex regulatory frameworks, technical specifications, and safety protocols. This is particularly acute for the domain of **Batteries and Accumulators**, a waste category that sits at the intersection of hazardous materials handling, producer responsibility legislation, and critical fire safety engineering. For the "Fessi" Knowledge Graph project, the integration of battery data is not merely a classification exercise; it is a safety-critical requirement mandated by the evolving landscape of European and German law, specifically the German Battery Act (BattG) and the overarching EU Battery Regulation 2023/1542.

This comprehensive research report documents the findings of an exhaustive data source discovery process aimed at populating the three layers of the Fessi knowledge graph: Core Entities, Logic & Rules, and Infrastructure. The investigation has uncovered a fragmented but highly structured ecosystem of data sources. While no single "master database" exists that aggregates legal, physical, and chemical attributes for all batteries, a robust graph can be constructed by synthesizing data from regulatory registers, technical standardization bodies, safety testing institutes, and manufacturer disclosures.

The research identifies **Stiftung EAR** as the foundational source for legal entity resolution, providing the authoritative link between brand names and their regulatory classification (Portable, Industrial, Automotive). This is complemented by the **Bundesanstalt für Materialforschung und -prüfung (BAM)** database and **ADR** regulations, which supply the dangerous goods logic (UN Numbers) necessary to determine transport and handling risks. For physical identification, **IEC 60086** standards and manufacturer **Safety Data Sheets (SDS)** from market leaders like Energizer and Duracell offer the granular chemical and dimensional data required to distinguish a benign alkaline cell from a high-risk lithium thionyl chloride battery. Finally, the infrastructure layer is populated by local university directives (e.g., RWTH Aachen, Uni Heidelberg) and insurance guidelines (**VdS 3103**), which dictate the specific physical containment protocols—such as the "Red/Yellow/Green" bin system—essential for campus safety.[1]

The following sections provide a detailed analysis of these sources, evaluating their schema compatibility, data richness, and integration complexity to guide the ETL (Extract, Transform,

Load) development of the Fessi system.

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1. Project Context and Domain Analysis

1.1 The Imperative of Granular Battery Data

In the context of a German university, the waste stream "Batteries & Accumulators" represents a disproportionate risk relative to its volume. Unlike paper or packaging, where misclassification results in minor recycling inefficiencies, the misclassification of batteries—specifically lithium-based chemistries—can lead to catastrophic thermal events. The Fessi chatbot must therefore possess a "Logic & Rules" layer capable of distinguishing between chemically distinct items that may share similar form factors. For instance, a standard AA battery could be Alkaline (safe for general battery bins) or Lithium Iron Disulfide (requiring distinct handling protocols in bulk).

The regulatory environment further complicates this. The **BattG** (Batteriegesetz) and the new **EU Battery Regulation 2023/1542** impose strict producer responsibility rules that differentiate batteries not just by chemistry, but by application (e.g., "Industrial" vs. "Portable"). A knowledge graph that fails to capture these distinctions will not only provide poor advice but may lead to regulatory non-compliance for the university as a waste operator.[1]

1.2 Defining the Ontology Requirements

To successfully answer user queries such as "Where do I throw away this e-bike battery?", the Fessi graph must instantiate specific nodes and relationships. The research focused on finding sources that could populate the following schema elements:

Graph Layer	Required Nodes/Properties	Target Data Attributes
Core Entities	WasteItem, Brand, Chemistry, Size	Manufacturer Names, Brand mappings, IEC Codes (e.g., CR2032), Chemical composition (Li-ion, NiMH).
Logic & Rules	DisposalRule, Condition, HazardClass	UN Numbers (3480 vs 3090), Fire Risk levels, ADR Special Provisions (SV 188), State logic

		(Damaged/Swollen).
Infrastructure	Container, Facility, CollectionPoint	Bin types (Red/Yellow/Green), Locations (Campus Building X), Pickup schedules.

The search strategy prioritized sources that offer structured data (CSV, Tables) over unstructured text, although safety manuals (PDFs) were found to be unavoidable for extracting conditional logic.

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2. Regulatory & Legal Framework Sources

The foundation of any waste management graph in Germany must be the legal definitions that govern producer responsibility and waste classification. Without these, the graph cannot accurately assign "Disposal Rules."

2.1 Source Profile: Stiftung EAR (Stiftung Elektro-Altgeräte Register)

The **Stiftung EAR** serves as the "Common Authority" (Gemeinsame Stelle) for the implementation of the Electrical and Electronic Equipment Act (ElektroG) and the Battery Act (BattG) in Germany. It is the single source of truth for all producers who legally place batteries on the market. For the Fessi graph, this is the primary dataset for entity resolution regarding brands and their legal categories.[1]

Source Identification

- **Name:** Stiftung EAR - Verzeichnis registrierter Hersteller (Batterierecht)
- **URL:** https://www.ear-system.de/ear-verzeichnis/battghersteller
- **Provider:** Stiftung EAR (Entrusted by the German Environment Agency/UBA).

Data Format & Access

- **Format:** The public register is accessible via a web interface that supports dynamic filtering. Crucially, it offers export functionality in **CSV**, **XLSX**, **XML**, and **PDF** formats.
- **Access Method: Free (Public Domain)**. No registration is required to access or download the public lists.
- **Rate Limits:** There are no published API rate limits for the web interface, but the system is designed for human query and bulk export rather than high-frequency programmatic access.
- **Bulk Download:** Yes. Users can generate a full list of all registered producers or filter by

category (e.g., "Gerätebatterien") and export the entire result set.[8]

Coverage & Quality

- **Record Count:** The database contains records for thousands of producers, covering every legal battery brand in the German market. It is comprehensive for compliance but does not include "grey market" or unregistered brands.
- **Geographic Scope: Germany**. This is the national implementation of the EU Directive.
- **Language:** German.
- **Update Frequency: Daily**. The register reflects the real-time status of producer registrations. If a manufacturer is deregistered, they disappear from the active list immediately.
- **Data Quality: Official Authority**. This data is legally binding. However, the categorization relies on the producers' self-declaration during the registration process.

Licensing

- **License Type:** Public administrative data. While not explicitly licensed under a Creative Commons deed, the data is published to fulfill a statutory transparency obligation.
- **Commercial Use:** Generally allowed for informational purposes, though redistribution of the raw database as a commercial product may be restricted. For a university knowledge graph, use is compliant.
- **Attribution:** Recommended to cite "Stiftung EAR" as the source of producer status.

Relevance Assessment

- **Coverage Relevance: 5/5 (Essential)**. This is the only way to reliably map a Brand Name (e.g., "Varta") to its Producer and legal battery categories.
- **Data Richness: 3/5**. The dataset is rich in legal metadata (Registration IDs, Brand Names, Categories) but completely lacks technical data (Chemistry, Dimensions, Voltage).
- **Accessibility: 5/5**. The CSV export feature makes ETL integration straightforward.
- **Overall Score: 5 (Essential)**.

Integration Notes

- **Useful Fields:**
  - Hersteller (Producer Name): The legal entity responsible.
  - Marke (Brand Name): The consumer-facing name printed on the battery.
  - Batt-Reg.-Nr. DE: The unique regulatory ID.
  - Batteriekategorie: The crucial classification field containing values like *Gerätebatterien* (Portable), *Industriebatterien* (Industrial), *Fahrzeugbatterien* (Automotive).[8]
- **Logic Mapping:**
  - The graph should use the Marke field to aliasing. If a user asks about "Bosch," the

graph queries the EAR nodes.
    - **Decision Rule:** If Brand is linked *only* to Industriebatterien in the EAR data, the chatbot can infer that the user is likely holding a specialized battery (e.g., for a power tool or e-bike) rather than a standard household battery. This triggers a specific disposal instruction: "This appears to be an industrial battery. Do not use the standard hallway bins."
- **Example Query:** A manual export filter can be applied for specific categories:
    - *Filter:* Batteriekategorie = "Industriebatterien" AND Marke = "Samsung".
    - *Result:* Confirms Samsung SDI's registration for industrial storage, distinct from their portable electronics division.

Gaps

The EAR register does not tell you *what* the battery is made of (e.g., Lithium vs. Lead-acid), only *who* made it and *what regulatory bucket* it falls into. It effectively covers the "Who" and "Legal What," but not the "Physical What."

2.2 Source Profile: EU Battery Regulation 2023/1542 Documentation

While not a database, the text of the new **EU Battery Regulation** serves as the definitive schema definition for the "WasteStream" and "LegalClassification" entities. It replaces the old directive and introduces new categories that must be reflected in the Fessi ontology.

Source Identification

- **Name:** Regulation (EU) 2023/1542 of the European Parliament and of the Council
- **URL:**(https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32023R1542)
- **Provider:** European Union / EUR-Lex.

Data Format & Access

- **Format:** PDF, HTML, XML.
- **Access Method:** Free.

Relevance & Integration

- **Ontology Updates:** The regulation introduces the **LMT Battery** (Light Means of Transport) category.[5] This is critical for a university setting where students use e-scooters and e-bikes.
    - *Old Schema:* Portable / Automotive / Industrial.
    - *New Schema:* Portable / SLI / LMT / EV / Industrial.
- **Integration Strategy:** Use this text to populate the Definition property of Category nodes.
    - *Node:* LMT Battery.
    - *Definition:* "Sealed, < 25kg, designed for traction of wheeled vehicles".[1]

- **Future Proofing:** The regulation mandates a "Battery Passport" (Batteriepass) for batteries > 2kWh and LMT batteries.[11] While the data for this is not yet fully available via public API, the Fessi graph should provision a property hasBatteryPassport (Boolean) to accommodate future QR-code scanning features.

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3. Dangerous Goods & Safety Logic Sources

The "Logic & Rules" layer of the graph is the brain of the chatbot. It determines whether a user should tape the poles of a battery, place it in a red bucket, or call for special pickup. This logic is derived from dangerous goods regulations.

3.1 Source Profile: BAM Dangerous Goods Database (DGG)

The **Bundesanstalt für Materialforschung und -prüfung (BAM)** is the supreme federal authority for hazardous materials in Germany. Their database is the gold standard for mapping battery types to **UN Numbers** and **ADR** (European Agreement concerning the International Carriage of Dangerous Goods by Road) classifications.

Source Identification

- **Name:** Datenbank GEFAHRGUT / DGG-Info
- **URL:** https://www.dgg.bam.de/quickinfo/de/
- **Provider:** BAM (Federal Institute for Materials Research and Testing).

Data Format & Access

- **Format:** Web Search Interface (HTML). The "QuickInfo" tool allows lookups by name or UN number.
- **Access Method: Free**.
- **Bulk Download:** No direct API or bulk download is advertised. Data must be extracted via specific queries or scraped (respecting robots.txt).
- **Rate Limits:** Standard web access limits apply.

Coverage & Quality

- **Record Count:** Covers the entire spectrum of dangerous goods (thousands of entries), including all battery-related UN numbers.
- **Geographic Scope:** International (UN Model Regulations) with specific notes for German/EU transport (ADR/RID).
- **Update Frequency:** Aligned with the biennial updates to the ADR (e.g., ADR 2023, ADR 2025).
- **Data Quality: Scientific Authority**. This is the highest level of data reliability for safety classifications.

Relevance Assessment

- **Coverage Relevance: 5/5**. Essential for determining the HazardRisk node.
- **Data Richness: 4/5**. Provides UN Number, Class, Packing Group, and special provisions (Sondervorschriften).
- **Accessibility: 2/5**. The lack of an API makes integration manual or scrape-dependent.
- **Overall Score: 4 (Essential Logic Source)**.

Integration Notes

- **Useful Fields:**
  - UN-Nummer: The key identifier (e.g., 3480).
  - Klasse: Almost always **Class 9** (Miscellaneous dangerous substances) for Lithium batteries.
  - Gefahrnummer: Hazard identification number (e.g., 90).
  - Sondervorschriften (Special Provisions): This is where the *conditional logic* lives. For example, **SV 376** applies to "Damaged/Defective Batteries" and mandates specific packaging.[2]
- **New Sodium-Ion Logic:** The research identified that **ADR 2025** introduces new UN numbers for Sodium-Ion batteries. The graph must be updated to include:
  - **UN 3551:** Sodium-ion batteries.
  - **UN 3552:** Sodium-ion batteries in equipment.[2]
  - *Logic:* "If Chemistry = Sodium-Ion -> Treat as UN 3551 -> Hazard: Similar to Lithium."
- **Mapping Strategy:** Create a lookup table in the graph:
  - Item: Lithium Ion Battery -> transportedAs -> UN 3480.
  - Item: Lithium Metal Battery -> transportedAs -> UN 3090.
  - Item: Damaged Lithium Battery -> transportedAs -> UN 3480 + Condition: Damaged -> Rule: SV 376.

3.2 Source Profile: VdS 3103 (Insurance Guidelines)

The **VdS 3103** leaflet, published by the German insurers (GDV), provides the *operational* logic for storing lithium batteries. While regulations tell you how to *transport* them, VdS tells you how to *store* them safely in a building (e.g., a university lab).

Source Identification

- **Name:** VdS 3103: Lithium-Batterien
- **URL:**(https://vds.de/fileadmin/Website_Content_Images/VdS_Publikationen/vds_3103_web.pdf)
- **Provider:** Gesamtverband der Deutschen Versicherungswirtschaft (GDV) / VdS Schadenverhütung GmbH.

Data Format & Access

- **Format:** PDF.
- **Access Method:** Free.

Relevance & Integration

- **Key Insight:** VdS 3103 categorizes lithium batteries by energy content, which directly translates to DisposalRule logic in the graph:
  - **Low Power:** < 100 Wh (Laptops, Phones) -> Rule: "Standard Collection".
  - **Medium Power:** > 100 Wh, < 12 kg (E-Bikes) -> Rule: "Separate storage, fire protection distance 2.5m".[15]
  - **High Power:** > 12 kg -> Rule: "Strict separation, consultation required".
- **Schema Enhancement:** The WasteItem node needs a property EnergyCapacity (Wh). The DisposalRule logic will query this property to select the correct VdS-compliant instruction.

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4. Technical & Identification Sources

Users typically describe batteries by what is written on them (e.g., "It says CR2032" or "It's an AA battery"). The graph needs to translate these physical identifiers into chemical definitions to assess risk.

4.1 Source Profile: IEC 60086 Standards

The **International Electrotechnical Commission (IEC)** standards define the alphanumeric nomenclature used on almost all batteries. This is the "Rosetta Stone" for battery codes.

Source Identification

- **Name:** IEC 60086 Primary Batteries
- **URL:** https://webstore.iec.ch/en/publication/23002 (Official Store) /(https://en.wikipedia.org/wiki/List_of_battery_sizes) (Accessible Data)
- **Provider:** IEC.

Data Format & Access

- **Format:** Official standards are Paid PDFs. Summaries and code tables are widely available in HTML/PDF format.
- **Access Method:** Paid (Official) / Free (Derivative tables).
- **Data Quality: High**.

Relevance & Integration

- **Coverage Relevance: 5/5**. Covers all standard consumer batteries.
- **Integration Strategy:** Construct a Nomenclature lookup table within the graph. The research highlights the following critical prefixes that map code to chemistry:

    - **R:** Zinc-Carbon (e.g., R6) -> Hazard: Low.
    - **LR:** Alkaline (e.g., LR6/AA) -> Hazard: Low.
    - **CR:** Lithium Manganese Dioxide (e.g., CR2032) -> Hazard: Fire Risk (Tape Poles).
    - **SR:** Silver Oxide -> Hazard: Resource Recovery.
    - **PR:** Zinc Air -> Hazard: Low.
    - **ER:** Lithium Thionyl Chloride -> Hazard: High (Toxic/Reactive).[17]
- **Richness:** The codes also define dimensions (e.g., in "CR2032", 20 = 20mm diameter, 32 = 3.2mm height). This allows the chatbot to help users identify "unknown" coin cells by asking them to measure the diameter.[18]

4.2 Source Profile: Manufacturer Safety Data Sheets (SDS)

For detailed chemical composition (essential for the "Material composition" requirement), Manufacturer SDSs are the best source.

Source Identification

- **Name:** Energizer / Duracell / Panasonic Technical Data Libraries
- **URL:** https://data.energizer.com/ [19], https://duracell.com/techlibrary/safety-data-sheets [20]
- **Provider:** Battery Manufacturers.

Data Format & Access

- **Format:** PDF.
- **Access Method:** Free.

Relevance & Integration

- **Data Richness: 5/5**. These documents provide the exact weight percentages of materials (e.g., "Lithium content: 0.03g"). They also list **Extinguishing Media** (e.g., "Use copious amounts of water for cooling" for Li-ion) which feeds the SafetyInstruction node.
- **Integration Challenge:** Requires PDF parsing.
- **Key Insight:** The SDSs confirm that modern alkaline batteries are generally mercury-free, allowing the graph to classify them as "Non-Hazardous Waste" (AVV 16 06 04) unless specific older date codes are found.[19]

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5. Visual Identification Sources

To support image-based queries ("What is this battery?"), the system needs training data.

5.1 Source Profile: Kaggle / TrashNet Datasets

Source Identification

- **Name:** Garbage Classification Dataset / TrashNet

- **URL:** https://www.kaggle.com/datasets/mostafaabla/garbage-classification [21]
- **Provider:** Community / Academic Researchers.

Data Format & Access

- **Format:** JPG Images.
- **Access Method:** Free.

Relevance & Integration

- **Data Quality: 3/5**. Most datasets have a generic "Battery" class. For Fessi, distinguishing *types* is critical.
- **Gap:** These datasets often lump all batteries together. They are useful for detecting "Is this a battery?", but poor for "Is this Lithium or Alkaline?".
- **Recommendation:** Use these to train a primary detector, but rely on text extraction (OCR) of the IEC codes (Source 4.1) for the actual classification logic.

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6. Infrastructure & Collection Sources

The final layer defines *where* the waste goes. This is highly localized.

6.1 Source Profile: University Disposal Handbooks (RWTH Aachen / Uni Heidelberg)

These documents are the "Ground Truth" for the specific operational rules of the campus.

Source Identification

- **Name:** Entsorgungshandbuch (Disposal Handbook)
- **URL:** https://www.rwth-aachen.de/global/show_document.asp?id=aaaaaaaaaazyhtp [4], https://www.uni-heidelberg.de/md/zentral/universitaet/beschaeftigte/service/sicherheit/bt-26.pdf [23]
- **Provider:** University Administration / Safety Officers.

Data Format & Access

- **Format:** PDF.
- **Access Method:** Free.

Relevance & Integration

- **Local Logic:** University rules often exceed municipal requirements.
  - *RWTH Aachen:* Mandates the use of the **SCA** (Service-Center Abfallwirtschaft) as the central intermediate storage. It categorizes waste into **H2** (Acute Toxic) and **E1** (Environmentally Hazardous) based on AVV codes.[4]

    - *Container Logic:* Identification of specific bin types. GRS defines a **Green/Yellow/Red** system:
        - **Green:** Standard portable batteries.
        - **Yellow:** High-energy Lithium (requires taping).
        - **Red:** Damaged/Defective Lithium (requires sand/pyrobubbles).[25]
- **Integration:** Map Building nodes to specific CollectionPoint instructions.
    - *Rule:* If Condition = "Damaged", Instruction = "Do not use building bin. Contact Safety Officer at Extension [Number] immediately."

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7. Integration Recommendations

To build the "Batteries" vertical of Fessi, the following ETL priority is recommended:

Phase 1: The Regulatory Skeleton (Stiftung EAR)

- **Action:** Ingest the Stiftung EAR CSV export.
- **Outcome:** A lookup table of every legal Producer and Brand in Germany, mapped to their broad LegalCategory (Portable vs. Industrial).
- **Value:** Immediate brand recognition and high-level sorting logic.

Phase 2: The Physical Decoder (IEC + Manufacturer Data)

- **Action:** Create a Nomenclature graph layer using IEC codes (CR, LR, SR). Parse key Manufacturer SDSs to link these codes to Chemistry and Hazard properties.
- **Outcome:** The ability to answer "What is a CR2032?" with "It is a Lithium Manganese Dioxide battery, 3V, Fire Risk."

Phase 3: The Safety Logic (BAM + VdS + University Rules)

- **Action:** Encode the **VdS 3103** power thresholds (<100Wh, >100Wh) into DisposalRule nodes. Map these rules to the specific **University Infrastructure** (e.g., "Yellow Bin in Building 4").
- **Outcome:** Context-aware safety advice that directs high-risk items to specialized facilities rather than general bins.

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8. Gaps Identified

1. **Real-Time State Detection:** No database can determine if a user's specific battery is *damaged*. The graph must rely on a "Diagnostic Dialog" (Logic Layer) to ask the user: "Is the casing swollen? Is it hot?" The *data* for the logic exists (VdS 3103), but the *instance data* is user-dependent.
2. **Product-Integrated Batteries:** There is no comprehensive open database linking

consumer products (e.g., "Singing Birthday Card") to their battery type. The graph must use heuristic rules (e.g., "If Item = Electronic Toy -> Assume Battery -> Ask User to Check").

3. **Sodium-Ion Specifics:** While ADR 2025 creates UN numbers for Sodium-ion, practical disposal infrastructure (bins/recyclers) for this new chemistry is lagging. The graph may need a temporary rule to "Treat as Lithium" until specific university protocols are updated.
4. **Legacy Brands:** The EAR register tracks *active* producers. Defunct brands (common in waste) are missing. A fallback to Wikidata or crowd-sourced lists may be necessary for legacy entity resolution.

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9. Detailed Data Tables

Table 1: IEC Prefix Mapping for Graph "Chemistry" Nodes

Prefix	Chemistry	Graph Hazard Flag	Nominal Voltage
R	Zinc-Carbon	Low	1.5V
LR	Alkaline	Low	1.5V
CR	Lithium Manganese Dioxide	**Fire Risk**	3.0V
SR	Silver Oxide	Low (Resource)	1.55V
PR	Zinc Air	Low	1.4V
ER	Lithium Thionyl Chloride	**Toxic / Fire**	3.6V
FR	Lithium Iron Disulfide	**Fire Risk**	1.5V

Table 2: UN Number Mapping for "Logic" Nodes

UN Number	Proper Shipping Name	Application in Graph

3480	Lithium Ion Batteries	Standard rechargeable Li-ion rule.
3481	Lithium Ion Batteries in Equipment	Logic for "Can I throw away the whole laptop?"
3090	Lithium Metal Batteries	Standard primary Lithium rule.
3091	Lithium Metal Batteries in Equipment	Logic for small electronics.
2794	Batteries, wet, filled with acid	Car/Lab lead-acid logic (Corrosive hazard).
3551	Sodium Ion Batteries (ADR 2025)	New logic path for emerging tech.

This report provides the blueprint for populating the Fessi knowledge graph with high-integrity, safety-compliant data. By synthesizing these diverse sources, the system can move beyond simple keyword matching to provide robust, context-aware disposal guidance that ensures regulatory compliance and campus safety.

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